STAT 20000 Chapter 26
Test of Significance
aka. Hypothesis Testing

Chapter 26: Test of Significance aka. Hypothesis Testing

We will show how to test whether the average of a box is a given value or not.

Jargons to learn:

- null hypothesis (H_0), alternative hypothesis (H_A)
- z-statistic
- P-value
- significance levels

Example: Flex-time

Flex-time: allowing employees to choose their working hours, like 8a-4p, 9a-5p, or 10a-6p

Example: Flex-time

Flex-time: allowing employees to choose their working hours, like 8a-4p, 9a-5p, or 10a-6p

A company introduces flex-time this year, wants to know if it changes absenteeism

- From past experience, average absenteeism $= 6.3$ days/year
- Management chooses a SRS of 100 employees to follow in detail. At the end of the year, these employees have an
  - average 5.4 days off from work, with an SD of 3.0 days.

Does flex-time made a real change in absenteeism?

Making the Question More Precise

- **Population**: all employees
- **Parameter**: the population average
  — the average absenteeism (days/year) of all employees
- **Sample**: the 100 selected employees
- **Statistic**: the sample average
  — the average days off of the 100 selected employees, $= 5.4$ days.

Does the average absenteeism change from 6.3 days?

Making the Question More Precise

- **Population**: all employees
- **Parameter**: the population average
  — the average absenteeism (days/year) of all employees
- **Sample**: the 100 selected employees
- **Statistic**: the sample average
  — the average days off of the 100 selected employees, $= 5.4$ days.

Does the average absenteeism change from 6.3 days?

Possible explanations:

- **Alternative hypothesis ($\mathbf{H_A}$)**: flex-time made a real change in absenteeism. (Parameter $\neq$ 6.3 days)

Making the Question More Precise

- **Population**: all employees
- **Parameter**: the population average
  — the average absenteeism (days/year) of all employees
- **Sample**: the 100 selected employees
- **Statistic**: the sample average
  — the average days off of the 100 selected employees, = 5.4 days.

Does the average absenteeism change from 6.3 days?

Possible explanations:

- **Alternative hypothesis ($\mathbf{H_A}$)**: flex-time made a real change in absenteeism. (Parameter $\neq$ 6.3 days)
- **Null hypothesis ($\mathbf{H_0}$)**: Average days off is still 6.3 days (Parameter = 6.3 days), and the difference is due to chance

A Box Model for the Flex-time Example

$$\boxed{??} \quad \boxed{??} \quad \boxed{??} \ldots$$

- Each employee in the company has 1 ticket in the box, showing .
- The sample is like ______ draws made at random __________ replacement from the box
- H_0: the average of the box $= 6.3$.
- H_A: the average of the box $\neq 6.3$.
- In plain words,
  - H_0 means flex-time made no change in the average absenteeism, and
  - H_A means flex-time made some change (up or down) in the average absenteeism.

A Box Model for the Flex-time Example

$$\boxed{??} \quad \boxed{??} \quad \boxed{??} \ldots$$

- Each employee in the company has 1 ticket in the box, showing the number of days that employee is absent.
- The sample is like ______ draws made at random __________ replacement from the box
- H_0: the average of the box $= 6.3$.
- H_A: the average of the box $\neq 6.3$.
- In plain words,
  - H_0 means flex-time made no change in the average absenteeism, and
  - H_A means flex-time made some change (up or down) in the average absenteeism.

A Box Model for the Flex-time Example

$$\underline{\big|\ \boxed{??}\quad \boxed{??}\quad \boxed{??}\ldots\ \big|}$$

- Each employee in the company has 1 ticket in the box, showing the number of days that employee is absent.
- The sample is like ___100___ draws made at random ___without___ replacement from the box
- H_0: the average of the box $= 6.3$.
- H_A: the average of the box $\neq 6.3$.
- In plain words,
  - H_0 means flex-time made no change in the average absenteeism, and
  - H_A means flex-time made some change (up or down) in the average absenteeism.

Test Statistic

- If H_0 is true, the expected value of the sample average is _______ days.

Test Statistic

- If H_0 is true, the expected value of the sample average is __6.3__ days.
- We observed a sample average of _______ days.

$$\textbf{observed} - \textbf{expected} = 5.4 - 6.3 = -0.9 \text{ days.}$$

Is the difference big or small?

Test Statistic

- If H_0 is true, the expected value of the sample average is $\underline{\quad 6.3 \quad}$ days.
- We observed a sample average of $\underline{\quad 5.4 \quad}$ days.

$$\textbf{observed} - \textbf{expected} = 5.4 - 6.3 = -0.9 \text{ days.}$$

Is the difference big or small?

- Let's compare the difference with the **SE**:

$$\begin{aligned}&\text{(estimated) SE of the sample average}\\ &= \frac{\text{estimated SD of box}}{\sqrt{\text{number of draws}}} = \frac{3.0}{\sqrt{100}} = 0.3 \text{ days}\end{aligned}$$

Test Statistic

- If H_0 is true, the expected value of the sample average is __6.3__ days.
- We observed a sample average of __5.4__ days.

$$\textbf{observed} - \textbf{expected} = 5.4 - 6.3 = -0.9 \text{ days.}$$

Is the difference big or small?

- Let's compare the difference with the **SE**:

$$\text{(estimated) SE of the sample average}$$
$$= \frac{\text{estimated SD of box}}{\sqrt{\text{number of draws}}} = \frac{3.0}{\sqrt{100}} = 0.3 \text{ days}$$

- The test statistic is the ratio of the difference (between the data and what's expected under H_0) and the SE, called the **z-statistic**

$$z = \frac{\text{observed} - \text{expected}}{\text{SE}} = \frac{-0.9 \text{ days}}{0.3 \text{ days}} = -3$$

Test Statistic (Cont'd)

For the hypotheses,

- H_0: the average of the box $= 6.3$;
- H_A: the average of the box $\neq 6.3$;

the bigger the *z*-**statistic** (regardless of sign),

- the more the data deviate from what's expected under H_0, and the more consistent with H_A.

P-value

The *P*-**value** is the **chance** of getting a result at least as deviant as the one observed, **assuming $\mathbf{H_0}$ is true**.

The smaller the *P*-value, the stronger the evidence against H_0.

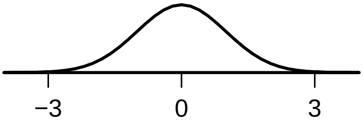
- Flex-time example: H_A says the average of the box $\neq 6.3$ days. Deviations from 6.3, either up or down, favor H_A over H_0. In terms of z, both large positive and large negative values argue against H_0.

P-value

The P-**value** is the **chance** of getting a result at least as deviant as the one observed, **assuming $\mathbf{H_0}$ is true**.

The smaller the P-value, the stronger the evidence against H_0.

- Flex-time example: H_A says the average of the box $\neq 6.3$ days. Deviations from 6.3, either up or down, favor H_A over H_0. In terms of z, both large positive and large negative values argue against H_0.
- If H_0 is right, the probability histogram for the z-statistic is approximately __normal__. The chance of getting a value of the z statistic at least as extreme (≥ 3 or ≤ -3) is about ______%.

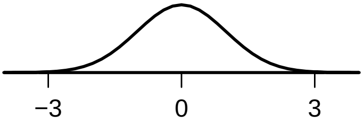


P-value

The P-**value** is the **chance** of getting a result at least as deviant as the one observed, **assuming $\mathbf{H_0}$ is true**.

The smaller the P-value, the stronger the evidence against H_0.

- Flex-time example: H_A says the average of the box $\neq 6.3$ days. Deviations from 6.3, either up or down, favor H_A over H_0. In terms of z, both large positive and large negative values argue against H_0.
- If H_0 is right, the probability histogram for the z-statistic is approximately __normal__. The chance of getting a value of the z statistic at least as extreme (≥ 3 or ≤ -3) is about __0.3__%.



P- value (Cont'd)

The smaller the P-value, the stronger the evidence against H_0, and the harder to believe H_0 is right.

- A P-value of $1/1000$ means that
  — if H_0 is true, only 1 in 1000 similar SRS would give a result at least as extreme as the one in hand.
  $\Rightarrow$ That's strong evidence against H_0.

P- value (Cont'd)

The smaller the P-value, the stronger the evidence against H_0, and the harder to believe H_0 is right.

- A P-value of $1/1000$ means that
  — if H_0 is true, only 1 in 1000 similar SRS would give a result at least as extreme as the one in hand.
  $\Rightarrow$ That's strong evidence against H_0.
- A P-value of $1/4$ means that,
  — if H_0 is true, 1 in 4 similar experiments would give a result at least as extreme as this one.
  $\Rightarrow$ No reason to disbelieve H_0.

P- value (Cont'd)

The smaller the P-value, the stronger the evidence against H_0, and the harder to believe H_0 is right.

- A P-value of $1/1000$ means that
  — if H_0 is true, only 1 in 1000 similar SRS would give a result at least as extreme as the one in hand.
  $\Rightarrow$ That's strong evidence against H_0.
- A P-value of $1/4$ means that,
  — if H_0 is true, 1 in 4 similar experiments would give a result at least as extreme as this one.
  $\Rightarrow$ No reason to disbelieve H_0.

The P-value is

- a measure of how surprising the observed data is, if H_0 is true;
- conversely, a measure of how plausible H_0 is, in the light of the observed data.

Conclusion of the Flex-time Experiment

The P-value 0.3% means that if the average days off per year hasn't changed from 6.3 days, only *3 in 1000* similar studies would have shown a greater apparent change.

- Average days off per year really did change.

The test of significance can only tell us whether there is a change.

But how big was the change in average days off for all employees in the company?

- 95%-confidence interval for the change is

Is is the change in average days off big enough use the Flex-time?

- That's question for management, not statistics.
- The test of significance can only tell us whether there is a change, but not whether the change is big enough to be important

Conclusion of the Flex-time Experiment

The P-value 0.3% means that if the average days off per year hasn't changed from 6.3 days, only *3 in 1000* similar studies would have shown a greater apparent change.

- Average days off per year really did change.

The test of significance can only tell us whether there is a change.

But how big was the change in average days off for all employees in the company?

- 95%-confidence interval for the change is $-0.9 \pm (2 \times 0.3) = -0.9 \pm 0.6$ days.

Is is the change in average days off big enough use the Flex-time?

- That's question for management, not statistics.
- The test of significance can only tell us whether there is a change, but not whether the change is big enough to be important

Conclusion of the Flex-time Experiment (Cont'd)

- Did flex-time cause the change?

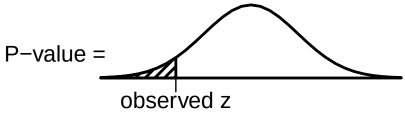
- Can you suggest a better experimental design?

One-Tailed Tests v.s. Two-Tailed Tests

If management believes that flex-time will not increase absenteeism, i.e. the average days off will either go down or remain the past value of 6.3 days but not go up, then H_A can be phrased as

$$\text{H}_A\text{: the average of the box } < 6.3.$$

In this case, only large negative values of the z-statistic are against H_0, and hence the P-value is the chance of getting a value of the z-statistic $\leq$ the observed



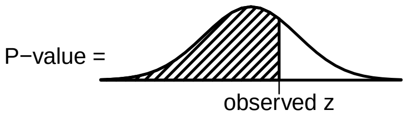
Tests with such alternatives are called **one-tailed** (or **one-sided**) tests, and the corresponding P-values are called **one-tailed** (or **one-sided**) P-values, as oppose to the two-tailed test and two-tailed P-value on page 9.

One-Tailed Tests v.s. Two-Tailed Tests (Cont'd)

For the one-tailed test with H_0 and H_A as follows,

$$\begin{aligned} \mathrm{H}_0 &: \text{ the average of the box} = 6.3, \\ \mathrm{H}_A &: \text{ the average of the box} < 6.3, \end{aligned}$$

if the z-statistic is positive, the P-value will then be at least 50%, which make sense since in that case, H_0 would be more likely to be true than H_A.

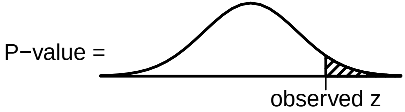


Upper One-Tailed Test

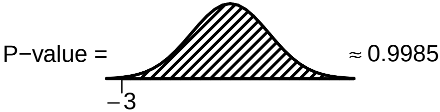
If management believes that flex-time might increase absenteeism but not decrease (which sounds counter-intuitive for this example), then H_A will be phrased as

$$H_A\text{: the average of the box is} > 6.3,$$

and the *P*-value is the chance of getting a value of the *z*-statistic $\geq$ the observed one.

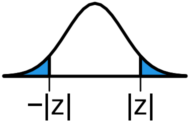
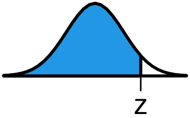
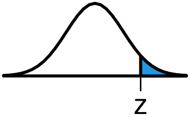
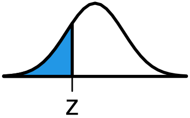
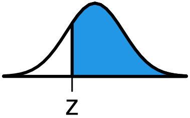


For the flex-time example, the observed *z*-statistic is -3.



Not surprisingly, a negative *z*-statistic means H_0 is more likely to be true than H_A.

Summary of the One-Tailed and Two-Tailed Tests

<table>
<tr><th></th><th>two−sided</th><th>lower
one−sided</th><th>upper
one−sided</th></tr>
<tr><td>H_0
H_A</td><td>Ave.of box $= \mu$
Ave.of box $\neq \mu$</td><td>Ave.of box $= \mu$
Ave.of box $< \mu$</td><td>Ave.of box $= \mu$
Ave.of box $> \mu$</td></tr>
<tr><td>P−value
when
z > 0</td><td></td><td></td><td></td></tr>
<tr><td>P−value
when
z < 0</td><td></td><td></td><td></td></tr>
</table>

Here μ is a given value, and z is the observed z-statistic.

Steps in Making a Test of Significance

- Formulate the *null hypothesis* (and perhaps the *alternative hypothesis* as statements about a box model for the data.
- Define a *test statistic* to measure the difference between the data and what's expected under H_0.
- Compute the *P-value* — the chance of getting a value for the test statistic as extreme, or more so, than the one observed.
- State your conclusion.

Example: Median Household Income In Atlanta

According to the 1999 census, the median household income in Atlanta (1.5 million households) was \$52,000. In June 2003, a market research organization takes a simple random sample of 750 households in Atlanta; 56% of the sample households had incomes over \$52,000. Did median household income in Atlanta increase over the period 1999 to 2003?

- Population: all households in Atlanta in 2003
- H_0: The median household income in Atlanta remained at \$52,000 in 2003
- H_A: The median household income in Atlanta was higher than \$52,000 in 2003

Example: Median Household Income In Atlanta (1)

Step 1: Set up a Box model for H_0:

- There is one ticket in the box for each
- What is the number on the ticket of a household shows?
  a. the household income for that household
  b. 1 if the household income for that household is over \$52,000 and 0 otherwise
- The H_0 says the sample is like 750 draws made at random from the box and the percentage of 1's in the box is ________. Options: 50%, 56%.
- The H_A says the percentage of 1's in the box is ____________.
- The percentage of 1's in the sample is ________. Options: 50%, 56%.

Example: Median Household Income In Atlanta (1)

Step 1: Set up a Box model for H_0:

- There is one ticket in the box for each households in Atlanta.
- What is the number on the ticket of a household shows?
  a. the household income for that household
  b. 1 if the household income for that household is over \$52,000 and 0 otherwise
- The H_0 says the sample is like 750 draws made at random from the box and the percentage of 1's in the box is ________. Options: 50%, 56%.
- The H_A says the percentage of 1's in the box is ____________.
- The percentage of 1's in the sample is ________. Options: 50%, 56%.

Example: Median Household Income In Atlanta (1)

Step 1: Set up a Box model for H_0:

- There is one ticket in the box for each households in Atlanta.
- What is the number on the ticket of a household shows?
  - a. the household income for that household
  - b. 1 if the household income for that household is over \$52,000 and 0 otherwise
- The H_0 says the sample is like 750 draws made at random from the box and the percentage of 1's in the box is __50%__. Options: 50%, 56%.
- The H_A says the percentage of 1's in the box is __over 50%__.
- The percentage of 1's in the sample is ________. Options: 50%, 56%.

Example: Median Household Income In Atlanta (1)

Step 1: Set up a Box model for H_0:

- There is one ticket in the box for each households in Atlanta.
- What is the number on the ticket of a household shows?
  a. the household income for that household
  b. 1 if the household income for that household is over $52,000 and 0 otherwise
- The H_0 says the sample is like 750 draws made at random from the box and the percentage of 1's in the box is __50%__. Options: 50%, 56%.
- The H_A says the percentage of 1's in the box is __over 50%__.
- The percentage of 1's in the sample is __56%__. Options: 50%, 56%.

Example: Median Household Income In Atlanta (2)

Step 2: Find the test statistic.

- The observed percentage of households with incomes over \$52K in the sample is ________

Example: Median Household Income In Atlanta (2)

Step 2: Find the test statistic.

- The observed percentage of households with incomes over \$52K in the sample is __56%__
- The expected percentage is ______ if the H_0 is right

Example: Median Household Income In Atlanta (2)

Step 2: Find the test statistic.

- The observed percentage of households with incomes over \$52K in the sample is __56%__
- The expected percentage is __50%__ if the H_0 is right
- What is the SE for the sample percentage?

$$\frac{\sqrt{0.56 \times 0.44}}{\sqrt{750}} \times 100\% \approx 1.813\% \text{ or } \frac{\sqrt{0.5 \times 0.5}}{\sqrt{750}} \times 100\% \approx 1.826\%$$

Example: Median Household Income In Atlanta (2)

Step 2: Find the test statistic.

- The observed percentage of households with incomes over \$52K in the sample is <u> 56% </u>
- The expected percentage is <u> 50% </u> if the H_0 is right
- What is the SE for the sample percentage?

$$\frac{\sqrt{0.56 \times 0.44}}{\sqrt{750}} \times 100\% \approx 1.813\% \text{ or } \frac{\sqrt{0.5 \times 0.5}}{\sqrt{750}} \times 100\% \approx 1.826\%$$

- The z-statistic is

$$z = \frac{\text{obs} - \text{exp}}{\text{SE}} \approx \frac{56\% - 50\%}{1.813\%} \approx 3.31 \text{ or } \frac{56\% - 50\%}{1.826\%} \approx 3.29.$$

Example: Median Household Income In Atlanta (3)

Step 3: Find the P-value.

- As H_A said the percentage is over 50%, we should calculate the upper one-sided P-value. From the normal table, the area correspond to $z = 3.29$ or $z = 3.31$ are both about 0.9995. So the upper one-sided P-value is

$$1 - 0.9995 = 0.0005.$$

Step 4: State your conclusion.

- In view of the small P-value, the H_0 looks implausible.

  The median income has increased.

Example: Simple Random Sample or Not

In a student project in the 1970s, a sample of 100 UC Berkeley students was taken. The sampling procedure is unclear but there were 53 men in the sample. From Registrar's data, 25,000 students were registered at Berkeley that term, and 67% were male.

Was the sample like a simple random sample?

Example: Simple Random Sample or Not (Cont'd)

Step 1: Set up a box model for the H_0.

- There is one ticket in the box for each ____________. Options:

Example: Simple Random Sample or Not (Cont'd)

Step 1: Set up a box model for the H_0.

- There is one ticket in the box for each ____________. Options:
  - person in the sample

Example: Simple Random Sample or Not (Cont'd)

Step 1: Set up a box model for the H_0.

- There is one ticket in the box for each ____________. Options:
  - person in the sample
  - student registered at Berkeley that term

Example: Simple Random Sample or Not (Cont'd)

Step 1: Set up a box model for the H_0.

- There is one ticket in the box for each ____________. Options:
  - person in the sample
  - student registered at Berkeley that term
- The ticket is marked 1 for the men and 0 for the women.

Example: Simple Random Sample or Not (Cont'd)

Step 1: Set up a box model for the H_0.

- There is one ticket in the box for each ____________. Options:
  - person in the sample
  - student registered at Berkeley that term
- The ticket is marked 1 for the men and 0 for the women.
- The number of tickets in the box is ____________ and the number of draws is ____________. Options: 100, 25,000.

Example: Simple Random Sample or Not (Cont'd)

Step 1: Set up a box model for the H_0.

- There is one ticket in the box for each ____________. Options:
  - person in the sample
  - student registered at Berkeley that term
- The ticket is marked 1 for the men and 0 for the women.
- The number of tickets in the box is ____________ and the number of draws is ____________. Options: 100, 25,000.
- The H_0 says that the sample is like _100_ draws made at random from the box.

Example: Simple Random Sample or Not (Cont'd)

Step 1: Set up a box model for the H_0.

- There is one ticket in the box for each ____________. Options:
  - person in the sample
  - student registered at Berkeley that term
- The ticket is marked 1 for the men and 0 for the women.
- The number of tickets in the box is ____________ and the number of draws is ____________. Options: 100, 25,000.
- The H_0 says that the sample is like __100__ draws made at random from the box.
- What does the H_A say?

Example: Simple Random Sample or Not (Cont'd)

Step 1: Set up a box model for the H_0.

- There is one ticket in the box for each ____________. Options:
  - person in the sample
  - student registered at Berkeley that term
- The ticket is marked 1 for the men and 0 for the women.
- The number of tickets in the box is ____________ and the number of draws is ____________. Options: 100, 25,000.
- The H_0 says that the sample is like __100__ draws made at random from the box.
- What does the H_A say?
- The percentage of 1's in the box is ____________.
  Options: 53%, 67%.

Step 2: Find the test statistic.

- The observed percentage in the sample is ________

Step 2: Find the test statistic.

- The observed percentage in the sample is __53%__
- The expected percentage is ______ if the H_0 is right

Step 2: Find the test statistic.

- The observed percentage in the sample is __53%__
- The expected percentage is __67%__ if the H_0 is right
- What is the SE for the number of men in the sample?

$$\frac{\sqrt{0.53 \times 0.47}}{\sqrt{100}} \times 100\% \approx 4.99\% \text{ or } \frac{\sqrt{0.67 \times 0.33}}{\sqrt{100}} \times 100\% \approx 4.70\%$$

Step 2: Find the test statistic.

- The observed percentage in the sample is __53%__
- The expected percentage is _67%_ if the H_0 is right
- What is the SE for the number of men in the sample?

$$\frac{\sqrt{0.53 \times 0.47}}{\sqrt{100}} \times 100\% \approx 4.99\% \text{ or } \frac{\sqrt{0.67 \times 0.33}}{\sqrt{100}} \times 100\% \approx 4.70\%$$

- The z-statistic is

$$z = \frac{\text{obs} - \text{exp}}{\text{SE}} \approx \frac{53\% - 67\%}{4.99\%} \approx -2.81 \text{ or } \frac{53\% - 67\%}{4.70\%} \approx -2.98.$$

Example: Simple Random Sample or Not (Cont'd)

Step 3: Find the P-value.

- Should we use one-side P-value or two-side P-value?

  In this case, large values of z regardless of sign indicate deviation from H_0. We should use two-side P-value.

  From the normal table, the area correspond to $z = -2.81$ (or $z = -2.98$) is about 0.0025 (or 0.0014). So the two-tailed P-value is

$$2 \times 0.0025 = 0.0050 \quad \text{or} \quad 2 \times 0.0014 = 0.0028.$$

Step 4: State your conclusion.

- In view of the small P-value, the H_0 looks implausible.

  The sample is unlikely to be a simple random sample.

Significance Levels

If the P-value of a result is less than $(100 \times \alpha)\%$,

- the result is said to be **significant at level** α, and
- H_0 is **rejected** at level α.

Significance Levels

If the P-value of a result is less than $(100 \times \alpha)\%$,

- the result is said to be **significant at level** α, and
- H_0 is **rejected** at level α.

e.g., if P-value $= 2.9\%$,

- the result is significant at the 3% level, and
- H_0 is rejected at level 0.03

Significance Levels

If the P-value of a result is less than $(100 \times \alpha)\%$,

- the result is said to be **significant at level** α, and
- H_0 is **rejected** at level α.

e.g., if P-value $= 2.9\%$,

- the result is significant at the 3% level, and
- H_0 is rejected at level 0.03

e.g., if P-value $= 3.1\%$,

- the result is significant at 5% level, but NOT at 3% level

Significance Levels

If the P-value of a result is less than $(100 \times \alpha)\%$,

- the result is said to be **significant at level** α, and
- H_0 is **rejected** at level α.

e.g., if P-value $= 2.9\%$,

- the result is significant at the 3% level, and
- H_0 is rejected at level 0.03

e.g., if P-value $= 3.1\%$,

- the result is significant at 5% level, but NOT at 3% level
- H_0 is rejected at level 0.05, but NOT at level 0.03

Significance Levels (Cont'd)

If P-value $< 5\%$, the result is called **statistically significant** (often shortened to **significant**).

- If H_0 is true, chance variation produces a significant result in about only 1 out of every 20 cases.
- If H_0 is true, H_0 is rejected at level 0.05 in about only 1 out of every 20 cases.

If P-value $< 1\%$, the result is called **highly significant**.

- If H_0 is true, chance variation produces a highly significant result in about only 1 out of every 100 cases.

True or False

- A "highly significant" result cannot possibly be due to chance.

- If a difference is "highly significant," there is less than a 1% chance for H_0 to be true.

.

- The P-value of a test is 1%, then the probability of the H_0 is true given the data is 1%.

.

- If the observed significance level is 5%, there are 95 chances in 100 for the H_A to be right.

.

True or False

- A "highly significant" result cannot possibly be due to chance. False. Even if H_0 is true, 1% of the time the experiment will give a result which is "highly significant."

- If a difference is "highly significant," there is less than a 1% chance for H_0 to be true.

.

- The P-value of a test is 1%, then the probability of the H_0 is true given the data is 1%.

.

- If the observed significance level is 5%, there are 95 chances in 100 for the H_A to be right.

.

True or False

- A "highly significant" result cannot possibly be due to chance.
  False. Even if H_0 is true, 1% of the time the experiment will give a result which is "highly significant."

- If a difference is "highly significant," there is less than a 1% chance for H_0 to be true.
  False. A *P*-value does not give the chance of H_0 being true. In fact, the *P*-value is computed assuming H_0 is true.

- The *P*-value of a test is 1%, then the probability of the H_0 is true given the data is 1%.

  .

- If the observed significance level is 5%, there are 95 chances in 100 for the H_A to be right.

  .

True or False

- A "highly significant" result cannot possibly be due to chance.
  False. Even if H_0 is true, 1% of the time the experiment will give a result which is "highly significant."

- If a difference is "highly significant," there is less than a 1% chance for H_0 to be true.
  False. A P-value does not give the chance of H_0 being true. In fact, the P-value is computed assuming H_0 is true.

- The P-value of a test is 1%, then the probability of the H_0 is true given the data is 1%.
  False.

- If the observed significance level is 5%, there are 95 chances in 100 for the H_A to be right.

  .

True or False

- A "highly significant" result cannot possibly be due to chance.
  False. Even if H_0 is true, 1% of the time the experiment will give a result which is "highly significant."

- If a difference is "highly significant," there is less than a 1% chance for H_0 to be true.
  False. A P-value does not give the chance of H_0 being true. In fact, the P-value is computed assuming H_0 is true.

- The P-value of a test is 1%, then the probability of the H_0 is true given the data is 1%.
  False.

- If the observed significance level is 5%, there are 95 chances in 100 for the H_A to be right.
  False.

Some Bad Ways to State the H_0 and H_A

- H_0: The average absenteeism is 6.3 days per year
- H_A: The average absenteeism is 5.4 days per year

or

- H_A: The average absenteeism for the 100 selected employees is 5.4 days per year

What's wrong: Hypotheses are statements about the population, not about the sample

Some Bad Ways to State the H_0 and H_A (Cont'd)

- H_0: The average absenteeism is not significantly reduced
- H_A: The average absenteeism is significantly reduced

What's wrong: "Significantly reduced" is an *uncertain* phrase. It means that from the data, the average absenteeism appears reduced, but we are not 100% sure.

Only definite and certain phrases will be used in the H_0 and the H_A.

Another Bad Way to State the H_0 and H_A

- H_0: The difference is due to chance
- H_A: The difference is real

Many students wrote this since FPP uses them to explain H_0 and H_A in plain words. Formally, we will not state H_0 and H_A in this manner.

What FPP means is

> *The difference in the **sample** average and the population average under the H_0 is due to chance.*

That is, under H_0, the **population averages** stay unchanged, but the **sample average** is different frm it because of the chance variation in the sampling process.

Summary

- A **test of significance** gets at the question of whether an observed result is real (H_A) or just a chance variation (H_0).
- A legitimate test of significance requires a **box model** for the data. The H_0 has to be translated into a statement about the box model. Usually, H_A does too.
- A z-**statistic** is used to measure the difference between the data and what is expected under H_0. The z-**test** uses the statistic

$$z = \frac{\text{observed} - \text{expected}}{\text{SE}}.$$

  The expected value and SE are computed on the basis of H_0.

Summary (Cont'd)

- The P-**value** is the chance of getting a value for the test statistic as extreme as, or more so, than the observed one. The chance is computed on the basis that H_0 is true. Therefore, the P-value *does not* give the chance of H_0 being right.
- **Small** P-values are evidence against H_0; they indicate that something besides **chance error** was operating to produce the observed result.